

ETH Macro-Event Reaction Study — One-Page Research Summary

Asset/Data: ETHUSDT Binance USD-M perpetual, 1-minute klines, UTC. Event-day data spanning **2023-2026** (Powell/Jackson Hole, NFP, CPI, PPI, FOMC). Built and run in a cost-aware, leakage-controlled backtest framework.

MOTIVATION

My original goal was to build a profitable **2025 news-driven (macro-event) straddle** on ETH for a trading-firm application — go market-neutral into scheduled releases and harvest the post-news move. Building it surfaced a key insight: a "straddle" replicated with a **symmetric long+short position on a linear perpetual is net-zero by construction** (the two legs cancel their own P&L and the position only bleeds fees) — unlike an *options* straddle, it has no long- volatility convexity. That realization reframed the project: rather than force a winning symmetric straddle, I turned it into a disciplined **event-reaction study** — which macro events move ETH in a *tradeable* (asymmetric) way, and whether any edge is real or just market beta / a single volatility regime. The asymmetric variants (ladder TP/SL, directional, momentum) were my attempt to turn that net-zero structure into something with an actual payoff.

FRAMEWORK & DATA DISCIPLINE

- Canonical UTC timestamp (`open_time_utc`); mandatory **Step-1 verification before every run**: checkpoints reconciled to raw 1m **100% on OHLCV**, `timestamp == event_time + offset`, 1440 candles/day, no duplicates. Runs **halt** on any missing event — never silently drop.
- Strategies fixed up front (pre-event long/short, post-event momentum, TP/SL straddle grid); no threshold tuning to fit results; fees/slippage modeled explicitly (0.05% fee + 0.05% slip).

EVENT SCREENING (2025)

Compared all event types on one footing and **screened out the noise**: - **PPI**: no tradeable reaction (whipsaw double-stops). **FOMC pre-long**: negative contributor. - **Ordinary Powell/Fed speeches**: nothing. **Jackson Hole**: large but a single event. - **CPI**: cleanest, broad-based reaction → became the focus.

2025 FINDING (IN-SAMPLE)

CPI pre-event long (enter -10m, exit +60m), post-May 2025, net of 0.2%/trade: **+9.5% cumulative, -1.08% max drawdown, 86% win (6/7)**. A direction-neutral **path-dependent TP/SL bracket** (TP +1.5% / SL -1.0% per leg): **11/11 positive after 4-leg fees**. *Note: a static symmetric long+short on a linear perp has no convexity and is net-zero; this positive result comes entirely from the path-dependent TP/SL execution, not from holding a true static straddle.*

MULTI-LAYER VALIDATION (THE CORE OF THE WORK)

Test	2025 (in-sample)	Out-of-sample / control	Verdict
Non-event-day baseline	CPI long mean at 97.5th pct of random 60-min holds	random non-CPI holds avg -0.16%	not pure beta
CPI long, 2023-2024 OOS	+9.5% (post-May)	+0.06% mean, median -0.27%, 42% win	does NOT replicate
CPI long, 2026 OOS	+9.5% (post-May)	+0.38% mean, 67% win (4/6)	partial — weak-positive
Path-dependent TP/SL bracket, 2023-24 / 2026 OOS	11/11 positive	15/24 / 4/6	does NOT fully replicate
Same-day short control	-11%	2023 -, 2024 -, 2026 -4.6% (0/6)	negative every year, 2023-2026
Robustness across exit horizons	+15/+30/+60/+120m all positive	OOS all ≈0 / median negative	edge not an exit-time artifact

Regime gradient (the key pattern). CPI long directional return tracks the prevailing volatility regime monotonically: **2023-24 ≈0 → 2026 weak-positive (+0.38%) → 2025 strong (+1.31%/event)**. The effect strengthens with the regime rather than appearing/vanishing randomly — itself evidence the dependence is on regime, not on a single lucky year. *The post-May split was an ex-ante regime call (ETH entering a higher-volatility, higher- participation phase after reclaiming ~½ of its prior high), not a PnL-optimized cut — the 2023-24 and 2026 controls were run precisely to test, not flatter, that call.*

FINAL, FALSIFIABLE CONCLUSION

ETH's tradeable reaction to CPI — both the directional edge and the "always ≥1.5% breakout" volatility property — is **regime-conditional, not a stable structural property of ETH-vs-CPI**. The directional edge **scales with the volatility regime** (≈0 in 2023-24, weak in 2026, strong in 2025) and holds across all exit horizons, while the **short side is negative in every year 2023-2026** — so CPI drives a *directionally upward-biased reaction whose magnitude depends on the regime*, not two-sided noise. The headline 2025 result is a genuine regime effect, **not universal alpha** — established by explicit multi-year out-of-sample falsification rather than in-sample curve-fitting.

ANTICIPATED QUESTIONS & DEFENSE

- "**Isn't post-May 2025 cherry-picked?**" — Post-May was an *ex-ante regime call* (ETH entering a higher-volatility / higher-participation phase), **not** a PnL-optimized cut. The 2023-24 and 2026 out-of-sample windows were run to *test* that call; the result (effect is regime-dependent, not universal) is reported as-is.
- "**You said a symmetric straddle is net-zero, yet the bracket is 11/11 positive — contradiction?**" — No: the static symmetric long+short has **no convexity** and is net-zero by construction. The positive result comes from **path-dependent TP/SL execution** (asymmetric exits), not from holding a true static straddle. (Renamed accordingly: *path-dependent TP/SL bracket*, not "straddle.")
- "**Isn't the short-side result just crypto risk-on / long-run beta?**" — This **does not claim a stable standalone alpha**. It shows CPI reaction days carry a **directional asymmetry** relative to (a) random non-event 60-min holds and (b) same-day short controls — a *characterization of event behavior*, not an alpha claim.

SKILLS DEMONSTRATED

Event-study design; rigorous data integrity/verification; leakage control; cost/slippage modeling; alpha-vs-beta separation via baselines and directional controls; **out-of-sample falsification**; honest characterization of edge boundaries and single-event/regime dependence.